SEMESTER: Winter, SESSION: 2024-25

Examination & Semester: **B. Tech., 2nd Sem (Common), End Semester**

Time: 2 Hours | Subject: **Digital Electronics (NECE102)** | Max Marks: 50

Answer all questions. (The right side numbers indicate marks.)

(Write all parts of a particular question in one place. Show the necessary steps whenever required.)

1. a) Implement the following function with a MUX: 5
   $F(A, B, C) = \Sigma\, m(1, 3, 5, 6)$.
   Choose A and B as select inputs.

   b) Design a full adder using a suitable decoder and logic gates. 5

2. a) Simplify the following Boolean expression using Boolean algebra (do not use K-MAP or Truth Table). Also, realize the simplified expression using a minimum number of NAND gates. 3+2

   $$F = (A \oplus B) \oplus (AB)$$

   b) A PN flip-flop has four states: clear to 0 (Reset), no change, complement, and set to 1, when inputs P and N are 00, 01, 10, and 11, respectively. For this PN flip-flop, draw the excitation table and derive the state (characteristics) equation. 2+3

3. a) Design a 3-bit synchronous binary Up-Down Counter using T flip-flops with detailed steps including state diagram, next-state table, flip-flop transition table and implementation using logic gates and flip-flops. 2+2+3+3

4. a) State the clear difference between a register and a counter. 4-bit 1

   b) Draw and explain the circuit diagram of a Parallel-In Serial-Out (PISO) shift register, highlighting the operation under shift and load conditions. 4

   c) Design a 4-bit Gray to Binary code converter with proper derivation and corresponding logic diagram. 5

5. a) Write the function of the following pins of 8085 microprocessor: 2

   ALE, Reset Out, HLDA, SOD

   b) After the execution of the following program, write the content of the Accumulator, memory 5590H, 5591H in hexadecimal number system. Also write the content of the carry, Auxiliary carry, and zero flag in binary form. 3

```
LXI H 5590 H
MVI A F0 H
MOV M, A
MVI A 0F H
ADD M
INR A
STA 5591 H
HLT
```

c) Write the content of memory locations 4FFF H and 5000 H (in hex code) after the execution of the following program. **2**

MVI B 23H

MVI C 34H

LXIH 4332H

DAD B

MOV A, H

STA 5000H

MOV A, L

STA 4FFFH

HLT

d) Write the content of memory location 2500H, registers A and B in hex code, after the execution of the following program. **3**

XRA A

MOV B, A

INR B

ORI 0F H

ADD B

STA 2500 H

CMA

HLT